Free balancing equations worksheet worksheet 16

Maths | 9th grade | Algebra

1. If 3x - 2 = 2x + 5, what is the value of x?

A) 3
B) 7
C) -7
D) 0

2. Simplify: 4(2x - 1) - 3(x + 3)

A) 5x - 13
B) x - 13
C) 5x + 9
D) x + 9

3. The sum of two consecutive integers is 37. What is the smaller integer?

A) 18
B) 19
C) 20
D) 21

4. If y = 3x + 7 and x = 4, what is the value of y?

A) 12
B) 19
C) 20
D) 21

5. What is the solution of the inequality 5x - 4 > 3x + 6?

A) x > 5
B) x < 5
C) x > 1
D) x < 1

6. Find the value of x if 2(x + 5) = 3(x - 2).

A) 16
B) 6
C) -6
D) -16

7. If the sum of x and twice y is 10 and y = 3, find x.

A) 4
B) -2
C) 2
D) 6

8. If f(x) = x^2 - 4x + 3, what is f(2)?

A) 3
B) 1

C) -1
D) 0

9. Solve for x: 4x/5 - 2 = x/3 + 1

A) 5
B) 15
C) 10
D) 20

10. What is the solution to the equation 3(x - 1) = 2(x + 4)?

A) -16
B) 10
C) 14
D) -2

Answer Key

1. A
Subtract 2x from both sides: x - 2 = 5. Then add 2 to both sides: x = 7.

2. A
Expand to get 8x - 4 - 3x - 9, then combine like terms: 5x - 13.

3. B
Let the integers be x and x+1. Then x + (x+1) = 37; 2x + 1 = 37; 2x = 36; x = 18.

4. C
Substitute x with 4: y = 3(4) + 7 = 12 + 7 = 19.

5. C
Subtract 3x from both sides: 2x - 4 > 6; Add 4 to both sides: 2x > 10; x > 5.

6. D
Expand: 2x + 10 = 3x - 6; Rearrange: 10 + 6 = 3x - 2x; x = 16.

7. B
Substitute y with 3 in x + 2y = 10: x + 2(3) = 10; x + 6 = 10; x = 4.

8. D
Substitute x = 2 into f(x): f(2) = 2^2 - 4(2) + 3 = 4 - 8 + 3 = -1.

9. B
Clear fractions: Multiply through by 15: 12x - 30 = 5x + 15; 7x = 45; x = 45/7.

10. A
Expand: 3x - 3 = 2x + 8; Rearrange: 3x - 2x = 8 + 3; x = 11.